PurePy: A Functional Subset of Python

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PurePy is a pure functional subset of Python, providing a common functional language for use by researchers in programming languages and pedagogy, while providing a front-end familiar and accessible to the Python community. This specification defines a versioned standard with formal syntax, well-formedness rules based on a definite assignment analysis, and an operational semantics intended to preserve the meaning of well-formed PurePy programs in Python. By combining functional programming with Python-compatible surface syntax, PurePy aims to support portable implementations and lower-friction adoption of new language ideas for modelling, data processing, analysis, and visualisation.

1 Introduction

- Define subset of Python called PurePy, defined as a syntactic subset further restricted by a well-formedness predicate
- Define a semantics for PurePy such that the semantics of a PurePy program is identical to its semantics in Python
- Some well-formedness and evaluation rules could be given by a desugaring, but for now we avoid having two separate syntaxes

Python interoperability is a key goal, but PurePy is also motivated by the desire to carve out a well-behaved discipline for scientific programming *within* Python: a sublanguage in which permissive Python idioms that invite confusion are tightened or excluded.

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2 Syntax

The syntax of the language is given in Figure 1 and Figure 2. A *block* is a non-empty sequence of statements; either a top-level statement, a `def` body, or a branch of a conditional. A *mutual region* $d_1 \dots d_{n+1}$ is a maximal contiguous non-empty sequence of `def` statements in a block: any non-`def` statement in that block, by definition, separates mutual regions. Maximality is not enforced by the grammar in Figure 1; the parser is responsible for grouping consecutive `def`s into a single region. The well-formedness rules treat each region as one mutually-recursive binding (see §2.4.5).

2.1 Contexts

Metavariables x, f, c range over *variables*; idiomatically f is used when the variable is expected to denote a function and c when it is expected to denote a class.

Definition 1 (Context)**.** *A context* Γ *(or* Δ*) is a finite map from variables to one of the following* context entries θ*:*

- *the variable's* definite assignment *status* $a \in \mathbb{B} = \{\mathsf{tt}, \mathsf{ff}\}$*;*
- *a* module reference*, for a fully-qualified module name* q*: either a* stub $\mathsf{Mod}(q)$*, or* loaded *reference* $\mathsf{Mod}(q, \Gamma)$*, which also carries the member context* Γ *of* q*;*
- *a* class entry C*, of the form* $\mathsf{Cls}_\Gamma(q.c, \vec{x}, c')$*, recording the declaring context* Γ*, the qualified name* $q.c$ *of the class, sequence of distinct field names* $\vec{x}$*, and base class name* c' *(or* $\bot$ *for no base).*

Module references arise from import statements. A module declares each class name at most once, so the qualified name of a class entry is the identity of the class. The definite-assignment terminology is borrowed from Java [7, §16]; tt indicates *definitely assigned* and ff *not definitely assigned* (or, equivalently, degenerate types "any" and "none"). It is useful to represent contexts as total functions on the set of all variables by extending the codomain with $\bot$ to indicate *undefined*. We write $\Gamma, x{:}\,\theta$ for the context that sends x to θ and otherwise agrees with Γ, define $\mathrm{dom}(\Gamma) = \{x \mid \Gamma(x) \neq \bot\}$, and write $\varnothing$ for the context where all variables are undefined. For a set of variables X, write $\mathsf{tt}(X)$ for the context $\{x{:}\,\mathsf{tt} \mid x \in X\}$; similarly $\mathsf{ff}(X)$.

Context operations. Contexts admit a non-commutative monoid of sequential composition $\Gamma \cdot \Delta$ ("override") and an idempotent commutative semigroup of parallel composition $\oplus$ ("merge"). Together these form a *right near-semiring* [28]: $\cdot$ right-distributes over $\oplus$ (Lemma 1), but left distributivity does not hold. Both operations are defined pointwise.

Definition 2 (Context override)**.**

$$\begin{aligned} \bot \cdot \theta &= \theta \cdot \bot = \theta \\ \theta \cdot \theta' &= \theta' \quad (\textit{otherwise}) \end{aligned}$$

Override is thus right-biased and has $\bot$ as both left and right unit.

Definition 3 (Context merge)**.** *For definite-assignment statuses* $a, a' \in \mathbb{B}$*:*

$$\begin{aligned} \bot \oplus \bot &= \bot \\ a \oplus \bot = \bot \oplus a &= \mathsf{ff} \\ a \oplus a' &= a \wedge a' \end{aligned}$$

Merge is otherwise undefined. Restricted to $\mathbb{B}$, it returns ff when exactly one argument is defined and the conjunction otherwise; so $\bot$ is not a unit.

Definition 4 (Context extend)**.**

$$\begin{aligned} \mathsf{Mod}(q,\Gamma) \lhd \mathsf{Mod}(q,\Gamma') &= \mathsf{Mod}(q,\Gamma \lhd \Gamma') \\ \mathsf{Mod}(q) \lhd \mathsf{Mod}(q,\Gamma) = \mathsf{Mod}(q,\Gamma) \lhd \mathsf{Mod}(q) &= \mathsf{Mod}(q,\Gamma) \\ \bot \lhd \theta = \theta \lhd \bot &= \theta \\ \theta \lhd \theta' &= \theta' \quad (\textit{otherwise}) \end{aligned}$$

Extend combines module references for the same module recursively, preferring loaded references to stubs, and is otherwise right-biased like override. It is used when processing import prefixes, where two imports rooted at the same package must combine rather than shadow.

Lemma 1 (Right distributivity)**.**

$$(\Delta \oplus \Delta') \cdot \Gamma = (\Delta \cdot \Gamma) \oplus (\Delta' \cdot \Gamma).$$

Left distributivity ($\Gamma \cdot (\Delta \oplus \Delta') = (\Gamma \cdot \Delta) \oplus (\Gamma \cdot \Delta')$) does not hold. If it did, the following two programs would be equivalent, since sequential composition with a preceding assignment could be distributed into the branches of a conditional. But they are not: the left program is ill-formed, because a variable assigned before a conditional does not remain definitely assigned if only some branches assign it (see also §2.4.2).

```
def f(b):
    x = 1
    if b:
        x = 2
    else:
        pass
    return x  # error
```

```
def f(b):
    if b:
        x = 1
        x = 2
    else:
        x = 1
        pass
    return x  # ok
```

Operators. The metavariable $\odot$ ranges over the supported binary operators and $\ominus$ over the supported unary operators, both enumerated in Figure 3. The grammar of Figure 2 is intentionally ambiguous on operator forms; disambiguation is by the precedence and associativity given in Figure 3, which follow Python.

2.2 Modules and programs

A PurePy *program* is a finite map $\mathcal{M}$ from *fully-qualified module names* (Figure 1) to *module bodies* t, of which one is a distinguished *main module* named `__main__`; here $\mathrm{name}(q)$ is the string literal naming q, so $\mathrm{name}(\texttt{__main__}) = \texttt{"__main__"}$. We identify a module with its fully-qualified name, writing "module q" for the module named q. A module body is a sequence of imports $\vec{\imath}$ followed by a sequence of statements $\vec{s}$; since imports are not statements, they can appear neither in blocks (and hence not in function bodies) nor after the first non-import statement. The well-formedness and evaluation rules check and evaluate the statements of each body $\mathcal{M}(q)$ with the assignment $\texttt{__name__} = \mathrm{name}(q)$

s	::=	`pass`	(Pass)
	|	$x = e$	(Assign)
	|	e	(ExprStmt)
	|	`assert` e	(Assert)
	|	`assert` e, e'	(AssertMsg)
	|	`return`	(ReturnNone)
	|	`return` e	(Return)
	|	`if` e `:` b `elif` e_1 `:` b_1 $\ldots$ `elif` e_n `:` b_n	(If)
	|	`if` e `:` b `elif` e_1 `:` b_1 $\ldots$ `elif` e_n `:` b_n `else` `:` b'	(IfElse)
	|	`match` e`:` `case` p_1`:` b_1 $\ldots$ `case` p_{n+1}`:` b_{n+1}	(Match)
	|	d_1 $\ldots$ d_{n+1}	(MutualRegion)
	|	`@dataclass` `class` c `:` ϕ	(Dataclass)
	|	`@dataclass` `class` $c(c')$ `:` ϕ	(DataclassExtend)
ι	::=	`import` q	(Import)
	|	`from` q `import` $x_1, \ldots, x_{n+1}$	(FromImport)
ϕ	::=	x_1 `:` `Any` $\ldots$ x_{n+1} `:` `Any`	(FieldList)
	|	`pass`	(EmptyFieldList)
b	::=	s_1 $\ldots$ s_{n+1}	(Block)
t	::=	ι_1 $\ldots$ ι_m s_1 $\ldots$ s_n	(ModuleBody)
q	::=	$x_1.x_2.\cdots.x_{n+1}$	(Name)
d	::=	`def` $f(x_1, \ldots, x_n)$ `:` b	(Def)

Figure 1: Modules and blocks

e	::=	x	(Var)
	|	ℓ	(Literal)
	|	$e(e_1, \ldots, e_n)$	(Call)
	|	$q(e_1, \ldots, e_n, x_{n+1}{=}e_{n+1}, \ldots, x_{n+m}{=}e_{n+m})$	(Constr)
	|	$\{x_1{:}e_1, \ldots, x_n{:}e_n\}$	(Dict)
	|	$e.x$	(Attribute)
	|	$e[e']$	(Subscript)
	|	$e \odot e'$	(BinOp)
	|	$\ominus e$	(UnaryOp)
	|	e_1 `if` e `else` e_2	(CondExpr)
	|	$[e_1, \ldots, e_n]$	(List)
	|	$(e_1, \ldots, e_n)$	(Tuple)
	|	$[e\ q_1\ \ldots\ q_n]$	(ListComp)
	|	`lambda` $x_1, \ldots, x_n$: e	(Lambda)
q	::=	`if` e	(Guard)
	|	`for` x `in` e	(Generator)
ℓ	::=	n	(Number)
	|	`True`	(True)
	|	`False`	(False)
	|	`None`	(None)
p	::=	ℓ	(PatLiteral)
	|	$-n$	(PatNegLit)
	|	x	(PatVar)
	|	_	(PatWildcard)
	|	$[p_1, \ldots, p_n]$	(PatList)
	|	$(p_1, \ldots, p_n)$	(PatTuple)
	|	$q(p_1, \ldots, p_n, x_{n+1}{=}p_{n+1}, \ldots, x_{n+m}{=}p_{n+m})$	(PatConstr)
	|	p `as` x	(PatAs)

Figure 2: Expressions and patterns

Level	Operators	Arity	Associativity
1 (lowest)	`or`	binary	left
2	`and`	binary	left
3	`not`	unary	—
4	`==` `!=` `<` `<=` `>` `>=`	binary	left
5	`+` `-`	binary	left
6	`*` `/` `//` `%`	binary	left
7	`+` `-`	unary	—
8 (highest)	`**`	binary	right

Figure 3: Operator precedence (lowest binds loosest); follows Python

prepended, modelling the `__name__` binding the runtime supplies automatically. $\mathcal{M}$ additionally always includes the predefined modules of Figure 4, each defining exactly its listed members, and is prefix-closed: the parent of any module in $\mathrm{dom}(\mathcal{M})$ is itself in $\mathrm{dom}(\mathcal{M})$.[1] The map $\mathcal{M}$ is fixed and ambient in all definitions that follow.

Definition 5 (Builtins context)**.** $\Gamma_{\texttt{builtins}}$ *binds the names of the predefined module* `builtins` *(Figure 4).*

Module	Members
`builtins`	`print`, `len`, `range`
`math`	
`sys`	`argv`, `exit`
`typing`	`Any`
`dataclasses`	`dataclass`

Figure 4: Predefined modules and their members

The metafunction assigns extends by union from individual statements to blocks, and imports and deps from individual imports to import sequences; on a module body $\vec{\imath}\ \vec{s}$, they act on $\vec{s}$ and $\vec{\imath}$ respectively. Write $p \preceq q$ (*prefix*) when $q = p$ or $q = p.q'$ for some q', and $\prec$ for the strict order.

The *dependency graph* $G_{\mathcal{M}}$ of a program has an edge (q, q') iff $q' \in \mathrm{deps}_q(\mathcal{M}(q))$. Every import depends on its target's ancestors; excepted are ancestors enclosing the importing module itself, which are necessarily still in progress when it runs and so cannot be required to load first. The exception appears only in the from-import clause because an import naming a proper descendant of the importing module is prohibited outright (§2.5). In the rules (Figure 12), the exemption is the bound on the loads-to judgement, whose prefixes the walk skips: ε for an import, the importing module for a from-import. Program well-formedness (Figure 13) requires the main module to be well-formed (which recursively requires its imports to be well-formed) and $G_{\mathcal{M}}$ to be acyclic.

2.3 Well-formedness

2.3.1 Auxiliary definitions

Three syntactic analyses support the well-formedness rules: $\mathrm{assigns}(s)$, the set of variables assigned anywhere in s (without descending into nested function definitions, which have their own scope, so that $\mathrm{assigns}(s) = \mathrm{dom}(\Delta)$ when $s : \mathsf{Assigns}\ \Delta$); $\mathrm{binds}(p)$, the set of variables introduced by a pattern p; and $\mathrm{captures}(s)$, the set of variables from the enclosing scope that are captured by closures (lambdas or nested function definitions) within s, with companion $\mathrm{captures}_{\mathsf{e}}(e)$ on expressions. All are defined in Figure 6. captures extends to blocks and module bodies by union (like assigns), with the singleton block identified with its statement.

Definition 6 (Pattern subsumption)**.** *The preorder $p \sqsubseteq p'$ (Figure 7) holds when every value matched by p is also matched by p'.*

[1] A source tree encodes such an $\mathcal{M}$: each module's body is its .py file, each package's body is its __init__.py, and a package directory without an __init__.py encodes the empty body (a namespace package, PEP 420, https://peps.python.org/pep-0420/). Prefix-closure thus holds by construction.

$$\begin{aligned}
\text{imports}(\texttt{import}\ q) &= \{q\} \\
\text{imports}(\texttt{from}\ q\ \texttt{import}\ x_1, \ldots, x_{n+1}) &= \{q\} \cup \{q.x_i \mid x_i \in \text{dom}(\text{submodules}(q))\}
\end{aligned}$$

$$\begin{aligned}
\text{deps}_q(\texttt{import}\ q') &= \text{imports}(\texttt{import}\ q') \cup \{p \mid p \prec q', p \neq q\} \\
\text{deps}_q(\texttt{from}\ q'\ \texttt{import}\ \vec{x}) &= \text{imports}(\texttt{from}\ q'\ \texttt{import}\ \vec{x}) \cup \{p \mid p \prec q', p \not\preceq q\}
\end{aligned}$$

$$\text{submodules}(q) = \{x{:}\ \mathsf{Mod}(q.x) \mid q.x \in \text{dom}(\mathcal{M})\}$$

(a) imports, deps, and submodules metafunctions

$$\begin{aligned}
\text{own-fields}(x_1 :\ \texttt{Any}\ \ldots\ x_{n+1} :\ \texttt{Any}) &= x_1, \ldots, x_{n+1} \\
\text{own-fields}(\texttt{pass}) &= \varepsilon
\end{aligned}$$

$$\begin{aligned}
\text{fields}(\mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot)) &= \vec{x} \\
\text{fields}(\mathsf{Cls}_\Gamma(q.c, \vec{x}, c')) &= \text{fields}(\Gamma(c')) \cdot \vec{x}
\end{aligned}$$

$$\begin{aligned}
\text{ancestors}(\mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot)) &= \{\mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot)\} \\
\text{ancestors}(\mathsf{Cls}_\Gamma(q.c, \vec{x}, c')) &= \{\mathsf{Cls}_\Gamma(q.c, \vec{x}, c')\} \cup \text{ancestors}(\Gamma(c'))
\end{aligned}$$

(b) own-fields, fields, and ancestors metafunctions

Figure 5: Module-related metafunctions

$$
\begin{aligned}
\text{assigns}(\texttt{pass}) &= \varnothing \\
\text{assigns}(x = e) &= \{x\} \\
\text{assigns}(e) &= \varnothing \\
\text{assigns}(\texttt{assert } e, e') &= \varnothing \\
\text{assigns}(\texttt{return}) &= \varnothing \\
\text{assigns}(\texttt{return } e) &= \varnothing \\
\text{assigns}(\texttt{if } e_1 : \ b_1 \ \texttt{elif } e_2 : \ b_2 \ \ldots \ \texttt{elif } e_{n+1} : \ b_{n+1}) &= \text{assigns}(\vec{b}) \\
\text{assigns}(\texttt{if } e_1 : \ b_1 \ \texttt{elif } e_2 : \ b_2 \ \ldots \ \texttt{elif } e_{n+1} : \ b_{n+1} \ \texttt{else} : \ b_{n+2}) &= \text{assigns}(\vec{b}) \\
\text{assigns}(\texttt{def } f_1(\vec{x}_1) : \ b_1 \cdot \ldots \cdot \texttt{def } f_{n+1}(\vec{x}_{n+1}) : \ b_{n+1}) &= \{f_1, \ldots, f_{n+1}\} \\
\text{assigns}(\texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots \ \texttt{case } p_{n+1}\texttt{: } b_{n+1}) &= \textstyle\bigcup_i \big(\text{binds}(p_i) \cup \text{assigns}(b_i)\big) \\
\text{assigns}(\texttt{@dataclass class } c : \ \phi) &= \{c\} \\
\text{assigns}(\texttt{@dataclass class } c(c') : \ \phi) &= \{c\}
\end{aligned}
$$

$$
\begin{aligned}
\text{binds}(\ell) &= \varnothing \\
\text{binds}(-n) &= \varnothing \\
\text{binds}(x) &= \{x\} \\
\text{binds}(_) &= \varnothing \\
\text{binds}([p_1, \ldots, p_n]) &= \textstyle\bigcup_i \text{binds}(p_i) \\
\text{binds}((p_1, \ldots, p_n)) &= \textstyle\bigcup_i \text{binds}(p_i) \\
\text{binds}(p \ \texttt{as } x) &= \text{binds}(p) \cup \{x\} \\
\text{binds}(q(p_1, \ldots, p_n, y_1{=}p_{n+1}, \ldots, y_m{=}p_{n+m})) &= \textstyle\bigcup_i \text{binds}(p_i)
\end{aligned}
$$

$$
\begin{aligned}
\text{captures}(\texttt{pass}) &= \varnothing \\
\text{captures}(x = e) &= \text{captures}_{\mathsf{e}}(e) \\
\text{captures}(e) &= \text{captures}_{\mathsf{e}}(e) \\
\text{captures}(\texttt{assert } e, e') &= \text{captures}_{\mathsf{e}}(e) \cup \text{captures}_{\mathsf{e}}(e') \\
\text{captures}(\texttt{return}) &= \varnothing \\
\text{captures}(\texttt{return } e) &= \text{captures}_{\mathsf{e}}(e) \\
\text{captures}(\texttt{if } e_1 : \ b_1 \ \texttt{elif } e_2 : \ b_2 \ \ldots \ \texttt{elif } e_{n+1} : \ b_{n+1}) &= \text{captures}_{\mathsf{e}}(\vec{e}) \cup \text{captures}(\vec{b}) \\
\text{captures}(\texttt{if } e_1 : \ b_1 \ \texttt{elif } e_2 : \ b_2 \ \ldots \ \texttt{elif } e_{n+1} : \ b_{n+1} \ \texttt{else} : \ b_{n+2}) &= \text{captures}_{\mathsf{e}}(\vec{e}) \cup \text{captures}(\vec{b}) \\
\text{captures}(\texttt{def } f_1(\vec{x}_1) : \ b_1 \cdot \ldots \cdot \texttt{def } f_{n+1}(\vec{x}_{n+1}) : \ b_{n+1}) &= \textstyle\bigcup_i \big(\text{fv}(b_i) \setminus (\{\vec{x}_i\} \cup \text{assigns}(b_i))\big) \\
&\quad \setminus \{f_1, \ldots, f_{n+1}\} \\
\text{captures}(\texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \ \ldots \ \texttt{case } p_{n+1}\texttt{: } b_{n+1}) &= \text{captures}_{\mathsf{e}}(e) \cup \\
&\quad \textstyle\bigcup_i \big(\text{captures}(b_i) \setminus \text{binds}(p_i)\big)
\end{aligned}
$$

$$
\begin{aligned}
\text{captures}_{\mathsf{e}}(x) &= \varnothing \\
\text{captures}_{\mathsf{e}}(\texttt{lambda } \vec{x}\texttt{: } e) &= \text{fv}_{\mathsf{e}}(e) \setminus \{\vec{x}\}
\end{aligned}
$$

Figure 6: Assigned, bound, and captured variables

$p \sqsubseteq p'$

sub-lit
$$\frac{}{\ell \sqsubseteq \ell}$$

sub-neg-lit
$$\frac{}{-n \sqsubseteq -n}$$

sub-var
$$\frac{}{p \sqsubseteq x}$$

sub-wild
$$\frac{}{p \sqsubseteq _}$$

sub-list-list
$$\frac{p_i \sqsubseteq q_i \quad (\forall i)}{[p_1, \ldots, p_n] \sqsubseteq [q_1, \ldots, q_n]}$$

sub-tuple-tuple
$$\frac{p_i \sqsubseteq q_i \quad (\forall i)}{(p_1, \ldots, p_n) \sqsubseteq (q_1, \ldots, q_n)}$$

sub-as-l
$$\frac{p \sqsubseteq q}{p \texttt{ as } x \sqsubseteq q}$$

sub-as-r
$$\frac{p \sqsubseteq q}{p \sqsubseteq q \texttt{ as } y}$$

Figure 7: Pattern p is subsumed by pattern p'

$\Gamma \vdash p_1 \cdot \ldots \cdot p_n$

pat-lit
$$\frac{}{\Gamma \vdash \ell}$$

pat-neg-lit
$$\frac{}{\Gamma \vdash -n}$$

pat-var
$$\frac{}{\Gamma \vdash x}$$

pat-wild
$$\frac{}{\Gamma \vdash _}$$

pat-list
$$\frac{\Gamma \vdash p \quad (\forall p \in \vec{p})}{\Gamma \vdash [\vec{p}]}$$

pat-tuple
$$\frac{\Gamma \vdash p \quad (\forall p \in \vec{p})}{\Gamma \vdash (\vec{p})}$$

pat-as
$$\frac{\Gamma \vdash p}{\Gamma \vdash p \texttt{ as } x}$$

pat-constr
$$\frac{\Gamma \vdash q \text{ names } C \qquad \text{fields}(C) = x_1, \ldots, x_n \cdot \vec{x}' \qquad \{\vec{y}\} = \{\vec{x}'\} \qquad \Gamma \vdash p_i \quad (\forall i)}{\Gamma \vdash q(p_1, \ldots, p_n, y_1{=}p_{n+1}, \ldots, y_m{=}p_{n+m})}$$

cases-nil
$$\frac{}{\Gamma \vdash \varepsilon}$$

cases-cons
$$\frac{\Gamma \vdash p \qquad \Gamma \vdash \vec{p}' \qquad \text{variables in } p \text{ distinct} \qquad p' \not\sqsubseteq p \quad (\forall p' \in \vec{p}')}{\Gamma \vdash p \cdot \vec{p}'}$$

Figure 8: Case patterns are well-formed under Γ

Subsumption is used by the well-formedness rule for `match` to rule out unreachable cases and to check exhaustiveness; with the current pattern set, the latter reduces to requiring a catch-all (var or wildcard) case.

2.3.2 Well-formedness rules

A pattern list is well-formed (Figure 8) when each pattern is linear (no repeated variable names) and no earlier pattern subsumes a later one. The `match` well-formedness rules use this judgement to enforce reachability and linearity of case patterns.

Every well-formed statement either *definitely returns*, or *definitely assigns* a context Δ, which may be empty. The base cases are that `return e` definitely returns, and `x = e` definitely assigns $\{\texttt{x}\colon \texttt{tt}\}$. *Result types* R classify these outcomes: R is either Returns (definitely returns) or Assigns Δ (definitely assigns Δ). Block well-formedness is given in Figure 9; expression well-formedness in Figure 10.

Context operations $\cdot$ and $\oplus$ lift to a right semiring over result types R, with Assigns $\varnothing$ as left and right unit for $\cdot$ and Returns as unit for $\oplus$ and zero for $\cdot$.

Definition 7 (Override and merge on result types)**.**

$$\begin{aligned}\mathsf{Assigns}\ \Delta \oplus \mathsf{Assigns}\ \Delta' &= \mathsf{Assigns}\ (\Delta \oplus \Delta')\\ R \oplus \mathsf{Returns} = \mathsf{Returns} \oplus R &= R\end{aligned}$$

$$\begin{aligned}\mathsf{Assigns}\ \Delta \cdot \mathsf{Assigns}\ \Delta' &= \mathsf{Assigns}\ (\Delta \cdot \Delta')\\ R \cdot \mathsf{Returns} = \mathsf{Returns} \cdot R &= \mathsf{Returns}\end{aligned}$$

The $\mathsf{Returns} \cdot R$ case never arises because the well-formedness judgement prohibits unreachable statements (§2.4.8).

Lemma 2 (Right distributivity (result types))**.**

$$(R \oplus R') \cdot S = (R \cdot S) \oplus (R' \cdot S)$$

A module attribute reference is well-formed only when it resolves to a definitely assigned member (rule attr-module); see §2.4.7.

The output context Γ of the module well-formedness judgement (Figure 13) comprises the classes and variables a module defines, together with its intrinsic `__name__` and a stub for each submodule, never the names it imports: following PEP 484[2], an imported name is a private dependency, not re-exported as part of the interface. Submodule stubs and the module's own bindings are disjoint, because the module rules prohibit a module from binding a name that is also the name of a submodule (§2.5.1).

2.4 Examples

2.4.1 Assignment as definition

Although PurePy is pure and all variable usage is well-scoped, local variables are introduced using an *assignment statement*, for compatibility with Python. If a variable is definitely assigned in any branch of a conditional, then for that variable to be usable after the conditional, every other branch of the conditional must either definitely return, or definitely assign that variable too. Thus the following are both well-formed:

```
if a < b:
   comp = "smaller"
else:
   comp = "bigger"
return comp
```

```
if a < b:
   return "smaller"
else:
   comp = "bigger"
return comp
```

The definite assignment set for a conditional that has at least one branch which definitely assigns is calculated by merging the definite assignment sets for all branches which definitely assign, using the $\oplus$ operator on environments.

Definite assignment thus gives each reference a unique lexical referent: a variable is introduced by an assignment, possibly jointly by the branches of a conditional, and a name may be referenced only

[2] https://peps.python.org/pep-0484/

$\Gamma \vdash b : R$

pass
$$\frac{}{\Gamma \vdash \texttt{pass} : \mathsf{Assigns}\ \varnothing}$$

assign
$$\frac{\Gamma \vdash e \qquad x \notin \mathrm{captures_e}(e)}{\Gamma \vdash x = e : \mathsf{Assigns}\ \{x : \mathsf{tt}\}}$$

expr-stmt
$$\frac{\Gamma \vdash e}{\Gamma \vdash e : \mathsf{Assigns}\ \varnothing}$$

assert
$$\frac{\Gamma \vdash e}{\Gamma \vdash \texttt{assert}\ e : \mathsf{Assigns}\ \varnothing}$$

assert-msg
$$\frac{\Gamma \vdash e \qquad \Gamma \vdash e'}{\Gamma \vdash \texttt{assert}\ e, e' : \mathsf{Assigns}\ \varnothing}$$

return-none
$$\frac{}{\Gamma \vdash \texttt{return} : \mathsf{Returns}}$$

return
$$\frac{\Gamma \vdash e}{\Gamma \vdash \texttt{return}\ e : \mathsf{Returns}}$$

if
$$\frac{\Gamma \vdash e_i \quad (\forall i) \qquad \Gamma \vdash b_i : R_i \quad (\forall i)}{\Gamma \vdash \texttt{if}\ e_1 :\ b_1\ \texttt{elif}\ e_2 :\ b_2\ \ldots\ \texttt{elif}\ e_{n+1} :\ b_{n+1} : R_1 \oplus \ldots R_{n+1} \oplus \ldots \oplus \mathsf{Assigns}\ \varnothing}$$

if-else
$$\frac{\Gamma \vdash e_i \quad (\forall i) \qquad \Gamma \vdash b_i : R_i \quad (\forall i)}{\Gamma \vdash \texttt{if}\ e_1 :\ b_1\ \texttt{elif}\ e_2 :\ b_2\ \ldots\ \texttt{elif}\ e_{n+1} :\ b_{n+1}\ \texttt{else} :\ b_{n+2} : R_1 \oplus \ldots \oplus R_{n+2}}$$

def
$$\frac{\Gamma \cdot \mathsf{tt}(\vec{f} \cdot \vec{x}_i) \cdot \mathsf{ff}(\mathrm{assigns}(b_i) \setminus \vec{x}_i) \vdash b_i : R_i \quad (\forall i) \qquad \vec{f}\ \mathrm{distinct}}{\Gamma \vdash \texttt{def}\ f_1(\vec{x}_1) :\ b_1 \cdot \ldots \cdot \texttt{def}\ f_{n+1}(\vec{x}_{n+1}) :\ b_{n+1} : \mathsf{Assigns}\ \mathsf{tt}(\vec{f})}$$

match
$$\frac{\begin{array}{c}\Gamma \vdash e \qquad \Gamma \cdot \mathsf{tt}(\mathrm{binds}(p_i)) \vdash b_i : R_i \quad (\forall i) \\ S_i = \mathsf{Assigns}\ \mathsf{tt}(\mathrm{binds}(p_i)) \cdot R_i \quad (\forall i) \qquad \Gamma \vdash \vec{p} \qquad p_{n+1} \in \{x, _\}\end{array}}{\Gamma \vdash \texttt{match}\ e\texttt{: case}\ p_1\texttt{:}\ b_1\ \ldots\ \texttt{case}\ p_{n+1}\texttt{:}\ b_{n+1} : S_1 \oplus \ldots \oplus S_{n+1}}$$

match-partial
$$\frac{\begin{array}{c}\Gamma \vdash e \qquad \Gamma \cdot \mathsf{tt}(\mathrm{binds}(p_i)) \vdash b_i : R_i \quad (\forall i) \\ S_i = \mathsf{Assigns}\ \mathsf{tt}(\mathrm{binds}(p_i)) \cdot R_i \quad (\forall i) \qquad \Gamma \vdash \vec{p} \qquad p_{n+1} \notin \{x, _\}\end{array}}{\Gamma \vdash \texttt{match}\ e\texttt{: case}\ p_1\texttt{:}\ b_1\ \ldots\ \texttt{case}\ p_{n+1}\texttt{:}\ b_{n+1} : S_1 \oplus \ldots \oplus S_{n+1} \oplus \mathsf{Assigns}\ \varnothing}$$

cons
$$\frac{\Gamma \vdash s : \mathsf{Assigns}\ \Delta \qquad \Gamma \cdot \Delta \vdash b : R \qquad \mathrm{captures}(s) \cap \mathrm{assigns}(b) = \varnothing}{\Gamma \vdash s \cdot b : \mathsf{Assigns}\ \Delta \cdot R}$$

Figure 9: Block b is well-formed in Γ

$\Gamma \vdash q$ names θ

simple-var
$$\frac{x\colon \mathsf{tt} \in \Gamma}{\Gamma \vdash x \text{ names } \mathsf{tt}}$$

simple-module
$$\frac{x\colon \mathsf{Mod}(q, \Gamma') \in \Gamma}{\Gamma \vdash x \text{ names } \mathsf{Mod}(q, \Gamma')}$$

simple-class
$$\frac{c\colon C \in \Gamma}{\Gamma \vdash c \text{ names } C}$$

qualified
$$\frac{\Gamma \vdash q \text{ names } \mathsf{Mod}(q', \Gamma') \qquad \Gamma' \vdash x \text{ names } \theta}{\Gamma \vdash q.x \text{ names } \theta}$$

(a) Name q resolves to context entry θ under Γ

$\Gamma \vdash e$

var
$$\frac{x\colon \mathsf{tt} \in \Gamma}{\Gamma \vdash x}$$

var-builtin
$$\frac{x \notin \mathrm{dom}(\Gamma) \qquad x\colon \mathsf{tt} \in \Gamma_{\texttt{builtins}}}{\Gamma \vdash x}$$

lambda
$$\frac{\Gamma \cdot \mathsf{tt}(\vec{x}) \vdash e}{\Gamma \vdash \texttt{lambda}\ \vec{x}\colon e}$$

constr
$$\frac{\Gamma \vdash q \text{ names } C \qquad \mathrm{fields}(C) = x_1, \ldots, x_n \cdot \vec{x}\,' \qquad \{\vec{y}\} = \{\vec{x}\,'\} \qquad \Gamma \vdash e_i \quad (\forall i)}{\Gamma \vdash q(e_1, \ldots, e_n, y_1{=}e_{n+1}, \ldots, y_m{=}e_{n+m})}$$

attr-module
$$\frac{\Gamma \vdash e.x \text{ names } \mathsf{tt}}{\Gamma \vdash e.x}$$

attr-object
$$\frac{\Gamma \vdash e}{\Gamma \vdash e.x}$$

(b) Expression e is well-formed under Γ

Figure 10: Expression well-formedness

$\Gamma, \vec{x}$ imports Γ'

imp-nil
$$\frac{}{\Gamma, \varepsilon \text{ imports } \varnothing}$$

imp-member
$$\frac{\Gamma \vdash x \text{ names } \theta \qquad \Gamma, \vec{x} \text{ imports } \Gamma'}{\Gamma, x \cdot \vec{x} \text{ imports } \{x\colon \theta\} \cdot \Gamma'}$$

imp-submodule
$$\frac{x\colon \mathsf{Mod}(q) \in \Gamma \qquad \vdash_{\mathsf{load}} q : \Delta \qquad \Gamma, \vec{x} \text{ imports } \Gamma'}{\Gamma, x \cdot \vec{x} \text{ imports } \{x\colon \mathsf{Mod}(q, \Delta)\} \cdot \Gamma'}$$

Figure 11: Names $\vec{x}$ imported from module context Γ produce bindings Γ'

$\vdash \vec{\iota} : \Gamma$

imports-nil

$$\frac{}{\vdash \varepsilon : \varnothing}$$

import

$$\frac{q' = x \cdot \vec{x} \qquad q \not\preceq q' \qquad \vdash_{\mathsf{load}} q' : \Delta \qquad \varepsilon, q', \mathsf{Mod}(q', \Delta) \text{ loads-to } \theta \qquad \vdash \vec{\iota} : \Gamma}{\vdash \texttt{import}\ q' \cdot \vec{\iota} : \{x : \theta\} \lhd \Gamma}$$

from-import

$$\frac{\vdash_{\mathsf{load}} q' : \Delta \qquad q, q', \mathsf{Mod}(q', \Delta) \text{ loads-to } \theta \qquad \Delta, \vec{x} \text{ imports } \Gamma' \qquad \vdash \vec{\iota} : \Gamma}{\vdash \texttt{from}\ q'\ \texttt{import}\ \vec{x} \cdot \vec{\iota} : \Gamma' \lhd \Gamma}$$

(a) Import prefix $\vec{\iota}$ of module q produces bindings Γ

$\vec{x}, q, \theta$ loads-to θ'

loads-to-simple

$$\frac{}{\vec{x}, x, \theta \text{ loads-to } \theta}$$

loads-to-qualified

$$\frac{q \not\preceq \vec{x} \qquad \vdash_{\mathsf{load}} q : \Gamma' \qquad \vec{x}, q, \mathsf{Mod}(q, \Gamma' \lhd \{x : \theta\}) \text{ loads-to } \theta'}{\vec{x}, q.x, \theta \text{ loads-to } \theta'}$$

loads-to-enclosing

$$\frac{q \preceq \vec{x}}{\vec{x}, q.x, \theta \text{ loads-to } \theta}$$

(b) Loaded reference θ for module q loads to the reference for the head of q

Figure 12: Import well-formedness

$\Gamma \vdash \vec{s} : \Delta$

nil

$$\frac{}{\Gamma \vdash \varepsilon : \varnothing}$$

statement

$$\frac{\Gamma \vdash s : \mathsf{Assigns}\ \Delta \qquad \Gamma \cdot \Delta \vdash \vec{s} : \Delta' \qquad \mathrm{captures}(s) \cap \mathrm{assigns}(\vec{s}) = \varnothing}{\Gamma \vdash s \cdot \vec{s} : \Delta \cdot \Delta'}$$

class

$$\frac{\nexists C'.\, c{:}\, C' \in \Gamma \qquad \text{own-fields}(\phi) = \vec{x}\ \text{distinct} \qquad C = \mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot) \qquad \Gamma, c{:}\, C \vdash \vec{s} : \Delta}{\Gamma \vdash \texttt{@dataclass class}\ c :\ \phi \cdot \vec{s} : \{c{:}\, C\} \cdot \Delta}$$

class-extend

$$\frac{\begin{array}{c}\nexists C'.\, c{:}\, C' \in \Gamma \qquad c'{:}\, \mathsf{Cls}_\Delta(q.c', \vec{y}, c'') \in \Gamma \\ \text{own-fields}(\phi) = \vec{x}\ \text{distinct} \qquad \vec{x} \cap \mathrm{fields}(\Gamma(c')) = \varnothing \qquad C = \mathsf{Cls}_\Gamma(q.c, \vec{x}, c') \qquad \Gamma, c{:}\, C \vdash \vec{s} : \Delta\end{array}}{\Gamma \vdash \texttt{@dataclass class}\ c(c') :\ \phi \cdot \vec{s} : \{c{:}\, C\} \cdot \Delta}$$

(a) Statements $\vec{s}$ of module q are well-formed in Γ

$\vdash_{\mathsf{load}} q : \Gamma$

module

$$\frac{\begin{array}{c}\mathcal{M}(q) = \vec{\iota}\,\vec{s} \qquad \vdash \vec{\iota} : \Gamma \\ \Gamma \vdash (\texttt{__name__} = \mathrm{name}(q)) \cdot \vec{s} : \Delta \qquad (\mathrm{dom}(\Gamma) \cup \mathrm{dom}(\Delta)) \cap \mathrm{dom}(\mathrm{submodules}(q)) = \varnothing\end{array}}{\vdash_{\mathsf{load}} q : \mathrm{submodules}(q) \cdot \Delta}$$

$\vdash \mathcal{M}$

program

$$\frac{\vdash_{\mathsf{load}} \texttt{__main__} : \Delta \qquad G_{\mathcal{M}}\ \text{acyclic}}{\vdash \mathcal{M}}$$

(b) Module and program well-formedness

Figure 13: Module well-formedness

where one such introduction covers every path. This is stricter than avoiding Python's `NameError`: the left-hand program of §2.1 assigns `x` on every run, but is rejected because the reference would denote the branch-local variable on one path and the variable it hides on the other.

2.4.2 Local assignment in conditional clauses

For flexibility, we permit *branch-local* variables. The assignment of `x` below (left) is permitted, and can be used locally in that branch of the conditional:

```
def foo():
  z = 6
  if b:
    x = 3
    y = x + 1
  else:
    y = 0
  return z
```

(a) Branch-local definition of `x`

```
def foo():
  x = 6
  if b:
    x = 3
    y = x + 1
  else:
    y = 0
  return x # error: partially defined
```

(b) Now with use of `x` after the conditional

However such variables only appear to be branch-local. The definition of $\oplus$ actually ensures that a variable assigned in any branch is considered assigned by the conditional; however unless it is assigned in *every* branch that does not definitely return, it will be bound to ff in the corresponding environment. Any attempt to refer to such a variable is illegal, as it refers to a variable which may not be assigned. (Thus the right-hand program is ill-formed, despite the enclosing `x = 6`, because in the `return x` statement, the inner assignment shadows the the outer one, and the inner assignment is only partial.) If we were to allow the program on the right, there would be no straightforward way to assign it a semantics which is both pure, and in agreement with Python, since in Python the function returns `3` in the case where `b` is `True`.

2.4.3 Function-level scoping of local variables

In Python, all local variables are scoped to the entire function body, not just from the point of assignment onwards. The $\mathrm{assigns}$ function in the mutual rule reflects this: all variables assigned anywhere in the function body are introduced at ff before the body is checked. This means the following program is statically rejected by PurePy, whereas Python rejects it at runtime with an `UnboundLocalError`:

```
y = 7

def f():
    x = y  # error: y is local (due to assignment below)
    y = 8

f()
```

Because `y` appears on the left-hand side of an assignment in `f`, it is in $\mathrm{assigns}(b)$ for the body of `f`, and is therefore introduced at ff at the top of the function. The reference to `y` in `x = y` then fails the well-formedness check, since $\Gamma(\mathtt{y}) = \mathsf{ff}$ rather than $\Gamma(\mathtt{y}) = \mathsf{tt}$.

2.4.4 Reassignment of captured variables

Once a variable has been captured by a closure, it cannot be reassigned in the enclosing scope, because its Python semantics would depend on mutation. Consider the following program:

```
def f():
    x = 5
    def g():
        return x
    x = 6
    return x + g()
```

In Python, `g` closes over the *mutable binding* of `x`, so `g()` returns `6` and the result is `12`. In a pure semantics where `x = 6` shadows the earlier binding, `g()` would return `5` and the result would be `11`. To preserve agreement with a subset of Python, we exclude this program from PurePy: the side condition in the seq rule requires that no variable captured by s is reassigned in the rest of the block b. Here `x` is captured by the definition of `g` and reassigned by `x = 6`, so the program is rejected. This is analogous to Java's requirement that local variables captured by a lambda must be *effectively final* [7, §15.27.2]: local variables are allocated on the stack, and so only the value of a local variable can be captured by a closure. A dataclass declaration counts as an assignment of its class name for these definitions, though re-declaring a class name is prohibited outright (§2.5).

The same issue arises when a variable is captured and reassigned in the *same* statement:

```
x = 5
def x():
    return x
x()  # Python: returns the function (late binding); pure: returns 5
```

The side condition on the assign rule rejects programs like this, requiring that the variable being bound is not captured by the expression. The mutual rule, by contrast, binds each f_i in the typing context for every body b_i, so the function names may legitimately appear in captures of those bodies. This is what permits self-recursion (the singleton-region case) and mutual recursion within a region.

2.4.5 Mutual recursion

The mutual rule binds the names $f_1, \ldots, f_{n+1}$ of a mutual region $d_1 \ldots d_{n+1}$ simultaneously. Each body b_i is checked in a context that includes all the f_j as definitely assigned, so any f_j may reference any other f_k within the region (the singleton case $n = 0$ permits self-recursion). The names in a region must be distinct.

Inserting any non-`def` statement between two `def`s splits them into separate regions, breaking the mutual reference:

```
def even(n):
    if n == 0:
        return True
    return odd(n - 1)

x = even(0)  # intervening statement separates the regions

def odd(n):
```

```
    if n == 0:
        return False
    return even(n - 1)
```

This program is rejected statically: `even` forms a singleton region in which `odd` is not yet bound, so the reference to `odd` in its body fails the var rule. Python accepts it: the call `even(0)` does not actually reach `odd`, and Python's late binding means `odd` is only looked up when needed.

Each branch of a conditional may contain its own mutual regions, subject to the usual definite assignment rules. The following is well-formed because both branches define both names:

```
b = True

if b:
    def f():
        return g()
    def g():
        return "via mutual region"
else:
    def f():
        return "f independent"
    def g():
        return "g independent"

print(f())
print(g())
```

A region containing two definitions of the same name is rejected directly by the mutual rule's distinct-names side condition:

```
def g():
    return 0

def f():
    return g()
def g():     # error: g defined twice in the same region
    return 1
```

(Python accepts this; late binding makes `f`'s call to `g` resolve to the second `def g`.)

Wrapping the second `def f; def g` pair in a conditional sidesteps the duplicate-name issue: the outer `def g` and the inner `def g` are now in different blocks, so they no longer fall in the same region. One might still expect a captured-variable problem, since the inner `def g` could be seen as rebinding the outer `g` that `f` captures. No such problem arises:

```
def g():
    return 0

if b:
    def f():
        return g()
    def g():
        return 1
```

The inner `def f; def g` form a branch-local mutual region, so `f`'s body resolves `g` to its sibling rather than to the outer `g`. The two are bound simultaneously, so `f` cannot be called before the inner `g` exists, and the value `f` sees is the same as the value Python's late binding finds at call time.

2.4.6 Pattern variables escape the match

A variable bound by a `case` pattern is in scope after the `match` statement, just as a variable assigned inside a conditional branch is in scope after the conditional. Python treats matching this way; PurePy follows.

```
v = 5
match v:
    case x:
        pass
print(x)    # prints 5
```

Here the catch-all `case x:` binds `x` in the body's scope, and the binding survives into the enclosing scope. Definite assignment after the match follows the same merge logic as a conditional: if every case binds the variable (or returns), it is definitely assigned afterwards; otherwise it is not (see also §2.4.2).

2.4.7 Uninitialised module members

In Python, an attribute of a module can be missing at runtime in two ways, both resulting in `AttributeError`; PurePy rejects both statically. First, the module body may never assign the member:

```
# lib.py                          # main.py
b = False                         import lib
if b:                             print(lib.x)  # AttributeError
    x = 1
```

Second, a submodule is not an attribute of its package until it is loaded (§2.5.1):

```
import a
a.b.f()  # AttributeError, unless a.b was loaded by an earlier import
```

Rule attr-module rejects these references: `x` is not definitely assigned in `lib`, and `a.b` is a stub, so neither resolves.

2.4.8 Unreachable code

A non-empty sequence of statements returns only if the last statement returns, and all preceding statements assign. Thus the following is illegal:

```
def foo(x):
  return x + 1
  return x    # error: unreachable
```

2.4.9 Conditionals without `else` clauses

An `if` statement without an `else` clause is equivalent to an `if` statement with an `else` clause whose body is a `pass` statement; such a statement never returns.

2.4.10 Implicit `None`

A `return` statement without an expression is equivalent to `return None`.

2.4.11 Implicit `return`

A function whose body s assigns rather than returns is equivalent to a function whose body is s followed by `return None`. This ensures every function returns, even if it is not explicitly specified.

2.4.12 `assert` without a message

An `assert e` statement without a message is equivalent to `assert e, ""`.

2.5 Prohibited Python features

The following Python features are excluded from PurePy as they are incompatible with purity or functional programming:

- `while` loops, `for` loops, `break`, `continue`
- `async`/`await`
- `global`, `nonlocal`
- `del`
- `try`/`except`/`raise`
- Augmented assignment (`+=`, `-=`, etc.)
- Walrus operator (`:=`)
- `yield`/`yield from`
- `class` definitions other than dataclasses
- `with` statements
- Truthiness: a non-boolean value used as a condition (the test of an `if` or conditional expression, or an `assert`). Python coerces any value to `bool`; PurePy requires the condition to be boolean.
- List patterns matching tuple values, or tuple patterns matching list values. In PurePy a list pattern matches only a list, and a tuple pattern matches only a tuple. Python permits either pattern against either kind of sequence.
- Positional class patterns matching fewer sub-patterns than the class's field arity. PurePy requires exact arity; Python permits any number from 0 up to the class's arity.
- Local or interleaved imports: an `import` or `from` statement inside a function or other block, or after a non-import statement. PurePy permits imports only at the start of a module; Python allows them anywhere.
- Wildcard imports (`from q import *`); PEP 8 recommends avoiding these[3].
- A submodule importing itself through its parent (`from a import b` inside module `a.b`). PurePy treats the from-imported submodule as a dependency, so this is an import cycle and is rejected; Python permits it, binding the partially-loaded module.
- An import (not a from-import) naming a proper descendant of the importing module (`import x.y` inside module `x`): the binding for the head would denote the still-loading module. Python permits it, binding the partially-initialised module. A from-import of the module's own submodule is permitted.

[3] https://peps.python.org/pep-0008/

- First-class classes: in Python, class declarations are assignments, so classes flow as values. In PurePy class names appear only at constructor and pattern positions, and declarations are module-level only.
- Re-declaring a class name: a dataclass declaration whose name already denotes a class in the module. Python permits the re-declaration, leaving the earlier class reachable only through its existing instances; PurePy requires a class name to be unique within its module, so that a qualified name identifies a class.
- First-class modules: in Python, modules are values. In PurePy a module name may be projected (member access) but not used as a value.
- A module binding a name that is also the name of one of its submodules (§2.5.1).
- Referring to a submodule the referring module does not itself import: `import a` followed by `a.b.f()` is rejected unless the module also imports `a.b`. Python permits the reference whenever some earlier import, anywhere in the program, loaded `a.b` (§2.4.7).

Complementing the list above (programs Python runs that PurePy rejects), a well-formed PurePy program avoids the Python runtime errors listed in Figure 16, under the conditions noted there. Errors from ill-typed operations, such as wrong operand types, calling a non-function, or attribute access on an object lacking the attribute, are outside the scope of well-formedness.

2.5.1 Variable and submodule with the same name

In Python, an attribute of a package can be bound by an assignment in the package's own body, and by a load of the same-named submodule, which assigns the new module to the attribute. Suppose the body of `pkg` is the single assignment `sub = 1`, and `pkg` also has a submodule `pkg.sub`:

```
import pkg                          import pkg.sub
print(pkg.sub)  # 1                 print(pkg.sub)  # <module 'pkg.sub'>
```

On the left, `pkg.sub` is never loaded, so the assignment is visible; on the right, the load overwrites the assignment. The name `pkg.sub` has no load-order-independent meaning, so the module rules prohibit a module from binding a name that is also the name of one of its submodules.

3 Operational semantics

Values, environments, and evaluation results are given in Figure 17. For a statement s, the result $\mathsf{assigns}\ \rho$ represents *normal completion* of s, with accumulated environment ρ; the result $\mathsf{returns}\ v$ represents *abrupt completion* of s with return value v. Composing statements sequentially composes their results according to the following partial monoid, analogous to sequential composition of result types:

$$\mathsf{assigns}\ \rho \cdot \mathsf{assigns}\ \rho' = \mathsf{assigns}\ (\rho \cdot \rho')$$
$$\mathsf{assigns}\ \rho \cdot \mathsf{returns}\ v = \mathsf{returns}\ v$$

with $\mathsf{assigns}\ \varnothing$ as left and right unit.

Module and program evaluation are given in Figure 21; import evaluation in Figure 20; importing names from a loaded module in Figure 19. Extend (Definition 4) extends to module values by reading its equations with a loaded reference $\mathsf{Mod}(q, \rho)$ carrying an environment in place of a context. The loading judgement $q \Rightarrow_{\mathsf{load}} \rho$ evaluates the import prefix of q and then its statements, mirroring the well-formedness judgement $\vdash_{\mathsf{load}} q : \Gamma$ (Figure 13); as with the static rules, the mutual recursion

Python error	Ruled out when	See
`NameError`, `UnboundLocalError`	Always: a variable may be referenced only where it is definitely assigned.	Figure 9, Figure 10
`ModuleNotFoundError`	Always: an import requires its target to be a module of the program.	Figure 12
`ImportError`	Always: a from-import requires each imported name to be a member of the source module, and import cycles are rejected. (A Python cycle may run or fail, depending on where the cycle is entered.)	Figure 11, Figure 13
`AttributeError`	Always: a module attribute reference must resolve to a definitely assigned member.	Figure 10
`TypeError`	The operation is a constructor call or class pattern, which must match the class's arity exactly. (Only excess sub-patterns are a Python error; a pattern with fewer is prohibited even though Python permits it.)	Figure 10, Figure 8

Figure 16: Python errors prevented by well-formedness

$$
\begin{array}{lcll}
v & ::= & \ell & \text{(Literal)} \\
 & | & [v_1, \ldots, v_n] & \text{(List)} \\
 & | & (v_1, \ldots, v_n) & \text{(Tuple)} \\
 & | & \mathsf{Lam}_\Gamma(\rho, \vec{x}, e) & \text{(LambdaClosure)} \\
 & | & \mathsf{Def}_\Gamma(\rho, \vec{d}, i) & \text{(DefClosure)} \\
 & | & \mathsf{Obj}_\Gamma(C, \rho) & \text{(Object)} \\
 & | & \mathsf{Mod}(q) & \text{(Module stub)} \\
 & | & \mathsf{Mod}(q, \rho) & \text{(Loaded module)} \\
 & | & \mathsf{Cls}_\Gamma(q.c, \vec{x}, c') & \text{(Class)} \\
\rho & ::= & \{x_1 \colon v_1, \ldots, x_n \colon v_n\} & \text{(Environment)} \\
r & ::= & \mathsf{assigns}\ \rho & \\
 & | & \mathsf{returns}\ v &
\end{array}
$$

Figure 17: Values, environments and evaluation results

$\vdash v$

$$\text{val-lit}\ \frac{}{\vdash \ell} \qquad \text{val-list}\ \frac{\vdash v_i \quad (\forall i)}{\vdash [v_1, \ldots, v_n]} \qquad \text{val-tuple}\ \frac{\vdash v_i \quad (\forall i)}{\vdash (v_1, \ldots, v_n)} \qquad \text{val-lambda}\ \frac{\vdash \rho : \Gamma \qquad \Gamma \cdot \mathsf{tt}(\vec{x}) \vdash e}{\vdash \mathsf{Lam}_\Gamma(\rho, \vec{x}, e)}$$

$$\text{val-def}\ \frac{\vdash \rho : \Gamma \qquad \Gamma \cdot \mathsf{tt}(\vec{f} \cdot \vec{x}_j) \cdot \mathsf{ff}(\mathrm{assigns}(b_j) \setminus \vec{x}_j) \vdash b_j : R_j \quad (\forall j) \qquad \vec{f}\ \text{distinct}}{\vdash \mathsf{Def}_\Gamma(\rho, \texttt{def}\ f_1(\vec{x}_1) :\ b_1 \cdot \ldots \cdot \texttt{def}\ f_{n+1}(\vec{x}_{n+1}) :\ b_{n+1}, i)} \qquad \text{val-obj}\ \frac{\vdash C \qquad \vdash \rho : \Gamma}{\vdash \mathsf{Obj}_\Gamma(C, \rho)}$$

$$\text{val-mod}\ \frac{q \in \mathrm{dom}(\mathcal{M})}{\vdash \mathsf{Mod}(q)} \qquad \text{val-mod-loaded}\ \frac{\vdash_{\mathsf{load}} q : \Gamma \qquad \vdash \rho : \Gamma}{\vdash \mathsf{Mod}(q, \rho)} \qquad \text{val-class}\ \frac{}{\vdash \mathsf{Cls}_\Gamma(q.c, \vec{x}, \bot)} \qquad \text{val-class-extend}\ \frac{c' : C' \in \Gamma}{\vdash \mathsf{Cls}_\Gamma(q.c, \vec{x}, c')}$$

$\vdash \rho : \Gamma$

$$\text{env-nil}\ \frac{}{\vdash \varnothing : \varnothing} \qquad \text{env-cons}\ \frac{\vdash v \qquad \vdash \rho : \Gamma}{\vdash \{x : v\} \cdot \rho : \{x : \theta\} \cdot \Gamma}$$

Figure 18: Well-formed values and environments

between loading and evaluation is grounded by the acyclicity of $G_{\mathcal{M}}$. Loading takes a single module; the ancestors of an imported module are loaded, once each, by the loads-to judgement.

The correspondence is exact: if $\vdash_{\mathsf{load}} q : \Gamma$ and $q \Rightarrow_{\mathsf{load}} \rho$ then $\vdash \rho : \Gamma$ (Figure 18). Since imports precede statements, every module a statement can reference is loaded before the statement runs, and since loading has no observable effects, reloading a module is unobservable; the semantics therefore needs no cache of loaded modules. These rules use the environment $\rho_{\texttt{builtins}}$ binding the names of Figure 4.

Block evaluation $\rho, b \Rightarrow r$ is given in Figure 22. Function definitions delimit the scope of sequential-composition effects, so as black boxes functions are always pure.

Expression evaluation $\rho, e \Rightarrow v$ is given in Figure 23. The pattern-matching judgement $\rho, p, v \rightsquigarrow \rho'$ (Figure 24) is a deterministic partial function; we write $\rho, p, v \not\rightsquigarrow$ (with no right-hand side) for $\nexists \rho'.\ \rho, p, v \rightsquigarrow \rho'$, i.e. matching is undefined. A constructor pattern matches any object whose class has the pattern's class among its ancestors; sub-patterns bind the fields of the pattern's class. An object records its class entry, whose qualified name is the identity of the class: since a module declares each class name at most once, the entry comparison in pat-constr amounts to a comparison of qualified names.

4 Related work

Subsets of programming languages have been defined for various purposes in the research literature:

Even though the syntax of programming languages if often described precisely, essentially none

$\rho, \vec{x}$ imports ρ'

imp-nil

$$\frac{}{\rho, \varepsilon \text{ imports } \varnothing}$$

imp-member

$$\frac{x{:}\, v \in \rho \qquad v \neq \mathsf{Mod}(q') \quad (\forall q') \qquad \rho, \vec{x} \text{ imports } \rho'}{\rho, x \cdot \vec{x} \text{ imports } \{x{:}\, v\} \cdot \rho'}$$

imp-submodule

$$\frac{x{:}\, \mathsf{Mod}(q') \in \rho \qquad q' \Rightarrow_{\mathsf{load}} \rho'' \qquad \rho, \vec{x} \text{ imports } \rho'}{\rho, x \cdot \vec{x} \text{ imports } \{x{:}\, \mathsf{Mod}(q', \rho'')\} \cdot \rho'}$$

Figure 19: Names $\vec{x}$ imported from module environment ρ produce bindings ρ'

$\vdash \vec{\iota} \Rightarrow \rho$

eval-imports-nil

$$\frac{}{\vdash \varepsilon \Rightarrow \varnothing}$$

eval-import

$$\frac{q' = x \cdot \vec{x} \qquad q' \Rightarrow_{\mathsf{load}} \rho \qquad \varepsilon, q', \mathsf{Mod}(q', \rho) \text{ loads-to } v \qquad \vdash \vec{\iota} \Rightarrow \rho'}{\vdash \texttt{import}\ q' \cdot \vec{\iota} \Rightarrow \{x{:}\, v\} \lhd \rho'}$$

eval-from-import

$$\frac{q' \Rightarrow_{\mathsf{load}} \rho \qquad q, q', \mathsf{Mod}(q', \rho) \text{ loads-to } v \qquad \rho, \vec{x} \text{ imports } \rho' \qquad \vdash \vec{\iota} \Rightarrow \rho''}{\vdash \texttt{from}\ q'\ \texttt{import}\ \vec{x} \cdot \vec{\iota} \Rightarrow \rho' \lhd \rho''}$$

(a) Import prefix $\vec{\iota}$ of module q produces bindings ρ

$\vec{x}, q, v$ loads-to v'

loads-to-simple

$$\frac{}{\vec{x}, x, v \text{ loads-to } v}$$

loads-to-qualified

$$\frac{q \not\preceq \vec{x} \qquad q \Rightarrow_{\mathsf{load}} \rho \qquad \vec{x}, q, \mathsf{Mod}(q, \rho \lhd \{x{:}\, v\}) \text{ loads-to } v'}{\vec{x}, q.x, v \text{ loads-to } v'}$$

loads-to-enclosing

$$\frac{q \preceq \vec{x}}{\vec{x}, q.x, v \text{ loads-to } v}$$

(b) Loaded reference v for module q loads to the reference for the head of q

Figure 20: Import evaluation

$\rho, \vec{s} \Rightarrow \rho'$

eval-nil

$$\frac{}{\rho, \varepsilon \Rightarrow \varnothing}$$

eval-statement

$$\frac{\rho, s \Rightarrow \mathsf{assigns}\ \rho' \qquad \rho \cdot \rho', \vec{s} \Rightarrow \rho''}{\rho, s \cdot \vec{s} \Rightarrow \rho' \cdot \rho''}$$

eval-class

$$\frac{C = \mathsf{Cls}_\Gamma(q.c, \text{own-fields}(\phi), \bot) \qquad \rho \cdot \{c\colon C\}, \vec{s} \Rightarrow \rho'}{\rho, @\texttt{dataclass}\ \texttt{class}\ c:\ \phi \cdot \vec{s} \Rightarrow \{c\colon C\} \cdot \rho'}$$

eval-class-extend

$$\frac{C = \mathsf{Cls}_\Gamma(q.c, \text{own-fields}(\phi), c') \qquad \rho \cdot \{c\colon C\}, \vec{s} \Rightarrow \rho'}{\rho, @\texttt{dataclass}\ \texttt{class}\ c(c'):\ \phi \cdot \vec{s} \Rightarrow \{c\colon C\} \cdot \rho'}$$

(a) Statements $\vec{s}$ of module q evaluate under environment $\rho : \Gamma$, producing bindings ρ'

$q \Rightarrow_{\mathsf{load}} \rho$

load

$$\frac{\mathcal{M}(q) = \vec{\iota}\,\vec{s} \qquad \vdash \vec{\iota} \Rightarrow \rho \qquad \rho, (\texttt{__name__} = \text{name}(q)) \cdot \vec{s} \Rightarrow \rho'}{q \Rightarrow_{\mathsf{load}} \text{submodules}(q) \cdot \rho'}$$

$\mathcal{M} \Rightarrow n$

eval-program

$$\frac{\texttt{__main__} \Rightarrow_{\mathsf{load}} \rho}{\mathcal{M} \Rightarrow 0}$$

(b) Module loading and program evaluation

Figure 21: Module and program evaluation

$$\begin{aligned}
\text{dispatch}(\rho, v, \varepsilon) &= (\varnothing, \texttt{pass}) \\
\text{dispatch}(\rho, v, (\texttt{case } p\texttt{: } b) \cdot \vec{c}) &= (\rho', b) && \text{if } \rho, p, v \rightsquigarrow \rho' \\
\text{dispatch}(\rho, v, (\texttt{case } p\texttt{: } b) \cdot \vec{c}) &= \text{dispatch}(\rho, v, \vec{c}) && \text{if } \rho, p, v \not\rightsquigarrow
\end{aligned}$$

$\rho, b \Rightarrow r$

eval-pass
$$\frac{}{\rho, \texttt{pass} \Rightarrow \textsf{assigns } \varnothing}$$

eval-assign
$$\frac{\rho, e \Rightarrow v}{\rho, x = e \Rightarrow \textsf{assigns } \{x\colon v\}}$$

eval-expr-stmt
$$\frac{\rho, e \Rightarrow v}{\rho, e \Rightarrow \textsf{assigns } \varnothing}$$

eval-assert
$$\frac{\rho, e \Rightarrow v \qquad v \neq \texttt{False}}{\rho, \texttt{assert } e \Rightarrow \textsf{assigns } \varnothing}$$

eval-assert-msg
$$\frac{\rho, e \Rightarrow v \qquad v \neq \texttt{False}}{\rho, \texttt{assert } e, e' \Rightarrow \textsf{assigns } \varnothing}$$

eval-return
$$\frac{\rho, e \Rightarrow v}{\rho, \texttt{return } e \Rightarrow \textsf{returns } v}$$

eval-def
$$\frac{d_i = \texttt{def } f_i(\vec{x}_i)\texttt{: } b_i \quad (\forall i) \qquad v_i = \textsf{Def}_\Gamma(\rho, \vec{d}, i) \quad (\forall i)}{\rho, \vec{d} \Rightarrow \textsf{assigns } \{\vec{f}\colon \vec{v}\}}$$

eval-if
$$\frac{\rho, e_j \Rightarrow \texttt{False} \quad (\forall j < i) \qquad \rho, e_i \Rightarrow \texttt{True} \qquad \rho, b_i \Rightarrow r}{\rho, \texttt{if } e_1\texttt{: } b_1 \texttt{ elif } e_2\texttt{: } b_2 \texttt{ } \ldots \texttt{ elif } e_{n+1}\texttt{: } b_{n+1} \texttt{ else: } b_{n+2} \Rightarrow r}$$

eval-else
$$\frac{\rho, e_j \Rightarrow \texttt{False} \quad (\forall j \in \{1, \ldots, n+1\}) \qquad \rho, b_{n+2} \Rightarrow r}{\rho, \texttt{if } e_1\texttt{: } b_1 \texttt{ elif } e_2\texttt{: } b_2 \texttt{ } \ldots \texttt{ elif } e_{n+1}\texttt{: } b_{n+1} \texttt{ else: } b_{n+2} \Rightarrow r}$$

eval-match
$$\frac{\rho, e \Rightarrow v \qquad \text{dispatch}(\rho, v, \texttt{case } p_1\texttt{: } b_1 \cdot \ldots \cdot \texttt{case } p_{n+1}\texttt{: } b_{n+1}) = (\rho', b) \qquad \rho \cdot \rho', b \Rightarrow r}{\rho, \texttt{match } e\texttt{: case } p_1\texttt{: } b_1 \texttt{ } \ldots \texttt{ case } p_{n+1}\texttt{: } b_{n+1} \Rightarrow r}$$

eval-cons
$$\frac{\rho, s \Rightarrow \textsf{assigns } \rho' \qquad \rho \cdot \rho', b \Rightarrow r}{\rho, s \cdot b \Rightarrow \textsf{assigns } \rho' \cdot r}$$

Figure 22: Block b evaluates to result r under environment $\rho : \Gamma$

$$\text{unwrap}(\textsf{returns}\ v) = v$$
$$\text{unwrap}(\textsf{assigns}\ \rho) = \texttt{None}$$

$$\text{env}(\rho, x) = \rho' \quad \text{where } \rho(x) = \textsf{Mod}(q', \rho')$$
$$\text{env}(\rho, q.x) = \rho' \quad \text{where } \text{env}(\rho, q)(x) = \textsf{Mod}(q', \rho')$$

$$\text{resolve-class}(\rho, c) = \rho(c)$$
$$\text{resolve-class}(\rho, q.c) = \text{env}(\rho, q)(c)$$

$\rho, e \Rightarrow v$

eval-var
$$\frac{x{:}\, v \in \rho}{\rho, x \Rightarrow v}$$

eval-var-builtin
$$\frac{x \notin \text{dom}(\rho) \qquad x{:}\, v \in \rho_{\texttt{builtins}}}{\rho, x \Rightarrow v}$$

eval-lambda
$$\frac{}{\rho, \texttt{lambda}\ x_1, \ldots, x_n{:}\ e \Rightarrow \textsf{Lam}_\Gamma(\rho, \vec{x}, e)}$$

eval-call-lambda
$$\frac{\rho, e \Rightarrow \textsf{Lam}_\Delta(\rho', \vec{x}, e') \qquad \rho, \vec{e} \Rightarrow \vec{u} \qquad \rho' \cdot \{\vec{x}{:}\, \vec{u}\}, e' \Rightarrow v}{\rho, e(e_1, \ldots, e_n) \Rightarrow v}$$

eval-call-def
$$\frac{\rho, e \Rightarrow \textsf{Def}_\Delta(\rho', \vec{d}, j) \qquad d_i = \texttt{def}\ f_i(\vec{x}_i) :\ b_i \quad (\forall i) \qquad v_i = \textsf{Def}_\Delta(\rho', \vec{d}, i) \quad (\forall i) \qquad \rho, \vec{e} \Rightarrow \vec{u} \qquad \rho' \cdot \{\vec{f}{:}\, \vec{v}\} \cdot \{\vec{x}_j{:}\, \vec{u}\}, b_j \Rightarrow r}{\rho, e(e_1, \ldots, e_n) \Rightarrow \text{unwrap}(r)}$$

eval-attr-module
$$\frac{\rho, e \Rightarrow \textsf{Mod}(q', \rho') \qquad x{:}\, v \in \rho'}{\rho, e.x \Rightarrow v}$$

eval-attr-object
$$\frac{\rho, e \Rightarrow \textsf{Obj}_\Delta(C, \rho')}{\rho, e.x \Rightarrow \rho'(x)}$$

eval-constr
$$\frac{\rho, e_i \Rightarrow v_i \quad (\forall i) \qquad \text{resolve-class}(\rho, q) = C \qquad \text{fields}(C) = x_1, \ldots, x_n, x_{n+1}, \ldots, x_{n+m}}{\rho, q(e_1, \ldots, e_n, y_1{=}e_{n+1}, \ldots, y_m{=}e_{n+m}) \Rightarrow \textsf{Obj}_\Gamma(C, \{x_1{:}\, v_1, \ldots, x_n{:}\, v_n, y_1{:}\, v_{n+1}, \ldots, y_m{:}\, v_{n+m}\})}$$

Figure 23: Expression e evaluates to value v under environment $\rho : \Gamma$

$\rho, p, v \rightsquigarrow \rho'$

pat-lit

$$\frac{}{\rho, \ell, \ell \rightsquigarrow \varnothing}$$

pat-neg-lit

$$\frac{}{\rho, -n, -n \rightsquigarrow \varnothing}$$

pat-var

$$\frac{}{\rho, x, v \rightsquigarrow \{x\colon v\}}$$

pat-wild

$$\frac{}{\rho, _, v \rightsquigarrow \varnothing}$$

pat-list

$$\frac{\rho, p_i, v_i \rightsquigarrow \rho_i \quad (\forall i)}{\rho, [p_1, \ldots, p_n], [v_1, \ldots, v_n] \rightsquigarrow \rho_1 \uplus \ldots \uplus \rho_n}$$

pat-tuple

$$\frac{\rho, p_i, v_i \rightsquigarrow \rho_i \quad (\forall i)}{\rho, (p_1, \ldots, p_n), (v_1, \ldots, v_n) \rightsquigarrow \rho_1 \uplus \ldots \uplus \rho_n}$$

pat-as

$$\frac{\rho, p, v \rightsquigarrow \rho'}{\rho, p \texttt{ as } x, v \rightsquigarrow \rho' \uplus \{x\colon v\}}$$

pat-constr

$$\frac{\begin{array}{c}\text{resolve-class}(\rho, q) = C \qquad C \in \text{ancestors}(C') \qquad \text{fields}(C) = x_1, \ldots, x_n, x_{n+1}, \ldots, x_{n+m} \\ \rho, p_i, \rho''(x_i) \rightsquigarrow \rho_i \quad (\forall i.\, 1 \leq i \leq n) \qquad \rho, p_{n+j}, \rho''(y_j) \rightsquigarrow \rho_{n+j} \quad (\forall j.\, 1 \leq j \leq m)\end{array}}{\rho, q(p_1, \ldots, p_n, y_1{=}p_{n+1}, \ldots, y_m{=}p_{n+m}), \mathsf{Obj}_{\Gamma'}(C', \rho'') \rightsquigarrow \rho_1 \uplus \ldots \uplus \rho_{n+m}}$$

Figure 24: Pattern p matches value v under environment ρ, producing bindings ρ'

of the languages in mainstream use today formally define their behaviour. Therefore, various well-defined subsets of programming languages were constructed to equip languages with formal (often even mechanised) semantics. Examples of this include work on Java [10], OCaml [25], JavaScript [5, 8, 20, 21], PHP [6], Rust [12, 13], and even lower-level languages and intermediate representations such as WebAssembly [29, 30, 31] and LLVM-IR [32, 33]. A particular success story of this line of work is the formalisation of a substantial subset of the C programming language, called Clight [4], which lead to the development of a verified compiler [17]. In a similar manner, the CakeML project [15] produced a verified implementation of a fragment of the ML language [18], based on previous efforts [16] of mechanising its semantics. Previous work on formalising the semantics of Python include an embedding [9] within the $\mathbb{K}$ semantics framework [24], a definition of structural operational semantics [14], and a reduction of most of the language to a core calculus [22].

A different line of work concerns the delineation of language subsets for a particular purpose. A prominent example is the *Source* family of languages [3], which are progressively extended subsets of JavaScript specifically geared towards introducing first-year computer science students to the foundations of computing. It is based on a new edition of a classic introductory text book [1], in which all program examples (which had previously been in Scheme) are illustrated in JavaScript. By successively increasing the allowed language fragment with each chapter (corresponding to different versions of the Source language), it allows to put focus on the particular aspects the authors are trying to convey, while disallowing students with prior knowledge of programming to take short-cuts by utilising more complex language features. The authors of [3] also remark, that the choice of JavaScript as a language was a pragmatic one: A previous study [26] showed that *industrial relevance* is perceived as the second-most important factor in choosing a programming language for introductory courses by educators in Australasia and the UK.

Likewise, various subsets of Python have been defined for different purposes in the past. ChocoPy [19] describes a subset specifically defined as a source language for the final project in courses on compiler construction, as most students are already familiar with the Python language by the time they take such a course. By restricting the language to a statically typed fragment, most student compilers can produce code that usually even outperforms the Python reference interpreter. In a similar manner, Restricted Python (RPython) [2] was defined as a strictly typed subset of Python, which removes some of the more dynamic features of the language. As part of the PyPy project [23], this allows RPython to be more easily compiled to common language runtimes, such as JVM and CLI/.NET.

We are aware of only one other work [27] specifically targetting a functional subset of Python, with the express intent of using it as a teaching tool. While we share a similar goal, our language fragment encloses more of the overall Python language, as their work doesn't allow for control structures such as conditionals or loops.

While it is not our goal to change Python towards a more functional programming language, it is interesting to notice that there are movements within the Python community for inclusion of programming principles commonly found in functional languages. An example of this is the inclusion of *lambda expressions* already in Python 1.0, and a more recent PEP draft [11] advocating for *deeply immutable objects*.

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